Deep Learning Report

1. Weights in Deep Learning

Weights are the core parameters of a neural network that determine how input data is transformed into output. Each connection between neurons has an associated weight, which represents the importance of that connection. During training, weights are adjusted using optimization algorithms such as gradient descent to minimize the loss function.

Initially, weights are randomly assigned. As training progresses, they are updated based on the error between predicted and actual values. Proper weight adjustment allows the model to learn patterns, relationships, and features from data.

2. Xception Algorithm

Xception stands for 'Extreme Inception'. It is a deep convolutional neural network architecture developed by François Chollet. It is based on depthwise separable convolutions, which split the convolution process into two parts: depthwise convolution and pointwise convolution.

In traditional convolution, spatial and cross-channel correlations are learned together. However, Xception separates these steps, making the model more efficient and powerful.

The architecture consists of entry flow, middle flow, and exit flow. Residual connections are also used to improve gradient flow and performance. Xception has been widely used in image classification and medical imaging tasks due to its high accuracy.

3. Model Saving in Deep Learning

Model saving is an essential practice in deep learning. It allows trained models to be stored and reused without retraining.

There are two main ways to save a model: saving the entire model (architecture + weights + optimizer state) or saving only the weights.

Saving models is useful for deployment, sharing, and resuming training. Popular frameworks like TensorFlow and PyTorch provide built-in functions such as model.save() and torch.save().

Model saving ensures reproducibility and helps in maintaining consistency across experiments.